Harpreet Singh

FPGA / HLS / Hardware Acceleration Engineer

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Summary

FPGA and embedded AI engineer focused on Vitis HLS, Vivado, MicroBlaze, Zynq, AXI4-Lite, AXI HP, BRAM data paths, fixed-point neural accelerators, and UART-visible board validation. Built research pipelines for QSNN, QiSNN, SNN, QLIF, PC-DDM-SNN, ANN baselines, N-MNIST, MNIST14, EEG intent decoding, rodent-raster decision decoding, and Zynq-based LLM acceleration.

Core Skills

- Xilinx Vivado, Vitis HLS, HLS C/C++, MicroBlaze, Zynq Cortex-A9, AXI4-Lite, AXI HP, BRAM maps, UARTLite
- Arty A7-35T, Arty A7-100T, Arty Z7-20, Artix-7, Zynq-7000, XSA, bitstream, Vitis platform, board bring-up
- Fixed-point arithmetic, ap_fixed<16,6>, ap_fixed<12,4>, quantized weights, INT8/INT4 matrix-vector acceleration
- ANN, SNN, QSNN, QiSNN, QLIF, PC-DDM-SNN, temporal N-MNIST, MNIST14, rodent raster, early-exit inference
- Python, NumPy, PyTorch, TensorFlow/Keras, model export, HLS testbenches, UART test automation, robustness evaluation

Professional Experience

Research Assistant, Core-Hub Neuroengineering Solutions | *University of Lethbridge* | *Apr 2025 - Present*

- Build FPGA research pipelines for ANN, SNN, QSNN/QiSNN, QLIF, PC-DDM-SNN, and rodent-raster decision experiments using Vitis HLS, Vivado, MicroBlaze, BRAM windows, and UART diagnostics.
- Developed dataset-to-BRAM flows using 196-feature MNIST14 inputs and 2352-word temporal N-MNIST/rodent-raster inputs with fixed-point payloads and software reference checks.
- Documented HLS/Vivado/Vitis rebuild flows, XSA/bitstream paths, UART tests, timing/resource evidence, and hardware-claim boundaries.

Research Assistant | *University of Regina* | *Jan 2021 - Apr 2024*

- Researched uncertain-reasoning intrusion detection for DoS/DDoS detection using Bayesian Networks, Markov Networks, Zeek, Wireshark, feature engineering, and Python ML workflows.
- Published SECRYPT 2024 work on uncertain-reasoning IDS methods and presented findings to an international cybersecurity research audience.
- Built ML prototypes including a Bayesian liver-disorder diagnostic model with 85% accuracy and Punjabi BERT/NLP classifiers for 100K+ news articles with 92% classification accuracy.

Independent Embedded Systems Developer | *Self-Employed* | *Jun 2024 - Mar 2025*

- Designed a custom STM32F405RGT6 MicroPython development board, including multilayer PCB design, firmware flashing, sensor/actuator interfaces, power management, and hardware/software debugging.

Senior Software Engineer | *Planetcast Media Services Ltd.* | *Dec 2018 - Nov 2020*

- Led a 15-member team delivering embedded hardware and firmware for broadcast/media systems, including a network-controlled 10-channel video switcher and Ethernet temperature acquisition platform.
- Designed schematics and PCBs, selected components, collaborated on cabinet/mechanical integration, and developed C firmware with Ethernet, HTTP/LWIP, alarm, and control interfaces.

Embedded Software Engineer / R&D Engineer | *Spark Eighteen Pvt. Ltd.; Exicom Tele-systems Ltd.* | *Feb 2016 - Dec 2018*

- Developed STM32/AVR embedded systems for LiFi control, telecom power plants, DC interface cards, and automatic test equipment.
- Designed 15-20 W isolated SMPS sections with 78-81% measured efficiency and performed board bring-up across CAN, SPI Flash, I2C EEPROM, Ethernet/LWIP, and GUI test automation.

Selected Projects

BCI_Game_Loop_FPGA

- Designed the FPGA/electronics branch of a closed-loop BCI stack on Arty A7-100T: ADS1299 frames feed MicroBlaze input BRAM, an EEG-adapted active temporal QiSNN produces five intent scores, and Ethernet/Wi-Fi/UART/SPI transports drive a PC/phone game with an idle reject state.

QSNN_EarlyExit_FPGA_Research

- Built an Arty A7-35T QSNN MicroBlaze system with AXI4-Lite control, BRAM input/output buffers, UART diagnostics, and MNIST board tests; recorded 0.698-0.699 ms HLS latency at 100 MHz, 9.87% LUTs, 5.13% FFs, 41.0% BRAM tiles, 3.33% DSPs, zero routing errors, and positive routed timing slack.

rodent_decision_qisnn_temporal

- Prepared a temporal QiSNN/SNN FPGA research pipeline for rodent decision prediction using DANDI-derived raster vectors, 12 time bins x 196 features, 2352-word Q10 input BRAM, 3-class lick labels, AXI4-Lite ap_ctrl_hs control, and debug-visible accelerator state planning.

Publication

- Harpreet Singh et al., "An Uncertain Reasoning-Based Intrusion Detection System for DoS/DDoS Detection," SECRYPT 2024, ISBN 978-989-758-709-2, ISSN 2184-7711, pp. 771-776.
- Published thesis: Singh H. A robust intrusion detection system utilizing uncertain reasoning techniques in artificial intelligence. The University of Regina (Canada); 2024.

Education

- PhD in Computational Behavioural Neuroscience, University of Lethbridge - In progress
- M.Sc. Computer Science, University of Regina - 2024
- Bachelor's equivalent in Electronics & Telecommunication Engineering, AMIETE/IETE - 2012